Binding Site Analysis — Nav1.5 N347K Mexiletine

fpocket Cavity Detection · Blind Docking · Targeted Docking · Comprehensive Score Summary

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1. Background and Aim

Nav1.5 N347K is a Brugada syndrome-associated mutation that suppresses sodium current. Mexiletine partially restores current in N347K (21% INa recovery, Frosio et al. 2023) but not in adjacent variants, suggesting a mutation-selective mechanism. Understanding where mexiletine binds in N347K and why it binds there selectively is essential for rational drug design.

Prof. Peri's request (meeting 20 April 2026): identify the best binding site(s) for mexiletine in Nav1.5 N347K, mapping all possible sites using multiple computational approaches.

This report documents the full binding site analysis pipeline: (1) druggable cavity detection with fpocket, (2) blind docking to validate receptor selection, (3) targeted docking at the top sites, and (4) integration with MD-derived MM-GBSA binding affinities.

2. Approach and Rationale

2.1. Why multiple methods?

No single method is sufficient for binding site identification in a large membrane protein like Nav1.5. Each method has limitations:

- fpocket detects geometric/chemical pockets on protein surface — fast, unbiased, but uses a single static frame
- Blind docking samples the entire protein surface — tests where a ligand prefers to go, but uses a rigid receptor
- Targeted docking focuses on specific sites with higher precision — better pose quality, but requires prior site knowledge
- MD + MM-GBSA captures dynamics and solvation — most accurate binding affinity, but computationally expensive and limited to simulated systems

By combining all four approaches, we achieve convergent evidence: a site confirmed by multiple independent methods is much more reliable than one identified by a single approach.

2.2. Why MD-derived representative frames?

A critical methodological question was whether to use static crystallographic structures (8VYK) or MD-derived conformations for fpocket and docking analyses. We tested both and found that static structures are insufficient for this system (see Section 3.3). The N347K substitution introduces a $K347^{+}$ charge that reshapes the protein's electrostatic environment over the course of MD simulation a conformational effect that is absent in the crystallographic structure, which was generated by simple in silico mutagenesis of a single residue.

All primary analyses therefore use the apoMutant MD representative frame (N347K, no ligand, index 3225 from 500 ns simulation, selected by ligand-filtered DBSCAN clustering).

2.3. Structures used

Structure	Type	Source	Used for
apoMutant MD rep (index 3225)	N347K, no ligand, 500 ns MD	DBSCAN medoid (RMSD 3.525 Å)	fpocket, blind docking, targeted docking
holoMutant MD rep (index 4835, ligand removed)	N347K + neutral mex, ligand removed	DBSCAN medoid	fpocket (induced fit), blind docking
N347K static PDB	8VYK + in silico N347K mutagenesis	PyMOL Wizard	fpocket comparison, blind docking
WT static PDB	8VYK, no mutagenesis	RCSB PDB	fpocket comparison, blind docking

Table 1. Receptor structures used in this analysis.

3. fpocket: Druggable Cavity Detection

fpocket v4.0 (Le Guilloux et al. 2009) was run on all four structures to detect and rank druggable cavities. The Druggability Score (0–1) reflects the likelihood that a pocket can bind a drug-like molecule.

3.1. apoMutant MD representative frame

134 pockets detected. Three showed pharmacologically relevant druggability:

Pocket	Drugg.	Volume (Å³)	PHE1760?	Key Residues & Location
2	**0.970 ★**	1312	**YES ✔**	PHE1760, TYR1409, THR1461, LEU1462, SER1458, TRP1345, ILE1756 — Domain IV S6 (classical LA site)
13	0.873	843	NO	ILE1630, LEU1627, ILE1633, LEU1634, CYS260, PHE259, TYR403 — Domain III S6 + Domain I S5-S6 (Domain III/IV junction)
8	0.747	497	NO	GLY780, TRP781, PHE784, LEU825, PHE1344 — Domain III S5-S6 (DIII-VSD allosteric)

Table 2. Top 3 pockets from apoMutant MD representative frame.

3.2. holoMutant MD rep (ligand removed) — induced fit

When the apoMutant and ligand-removed holoMutant conformations are compared, mexiletine binding clearly induces an expansion of the binding pocket (induced fit):

Pocket	Drugg.	Volume (Å³)	PHE1760?	Location
21	**1.000 ★**	2591	**YES ✔**	Domain IV S6 + Domain III/IV junction MERGED — PHE1760, TYR1767, TYR1409, LEU1717, LEU1721, VAL1400. After mexiletine binding, Pocket 2 and Pocket 13 merge into one extended cavity.
135	0.990	2017	NO	Domain III S6 fenestration — ARG1632, ILE1630, LEU1634, PHE1522. Opens specifically in holoMutant conformation. Consistent with tonic block entry route.

Table 3. Top pockets from holoMutant MD rep (ligand removed).

The Druggability Score increases from 0.970 (apoMutant) to 1.000 (holoMutant) and pocket volume doubles (1,312 → 2,591 Å³). This confirms an induced fit mechanism: mexiletine binding at Domain IV S6 reshapes the protein to create a larger, more druggable cavity extending toward Domain III/IV junction.

3.3. Static PDB structures — why they are insufficient

fpocket was also run on the N347K static PDB (8VYK + in silico N347K mutation) and the WT static PDB (8VYK) for comparison. The results were virtually identical between the two:

- N347K static: best pocket = Pocket 30 (druggability 0.975) — Domain III/IV inner helix. PHE1760 NOT present.
- WT static: best pocket = Pocket 30 (druggability 0.975) — same residues, same score as N347K static.

The results are identical because in silico mutagenesis changes only one residue (N347→K) without altering the overall protein conformation. The true effect of $K347^{+}$ on pocket geometry only emerges after 500 ns MD simulation, where the charge reshapes local electrostatics and opens the PHE1760-containing cavity. Static structures therefore underestimate the druggability of the primary binding site and cannot capture the mutation-specific conformational effect.

> PHE1760 appears in the top pocket only in MD-derived structures (apoMutant Pocket 2, holoMutant Pocket 21). It is absent from the top pocket in both static structures. This is a critical methodological finding.

4. Blind Docking: Receptor Selection Validation

Blind docking of protonated mexiletine (+1, pKa ≈9.1) was performed on all four receptor structures (AutoDock Vina, exhaustiveness = 128, 20 modes) to independently assess which receptor conformation correctly presents the binding site. The closest protein residues to Mode 1 (best pose) were identified by distance analysis.

Receptor	Score	Mode 1 Closest Residues	Site & Assessment
apoMutant MD rep (index 3225)	**−6.3 kcal/mol**	PHE1760 (2.97 Å) THR1461 (1.67 Å) LEU1462 (2.63 Å) TRP1345 (2.13 Å)	**Domain IV S6 ✔ Classical LA binding site confirmed Modes 1–5 all within 3 Å of same site**
N347K static PDB	−6.5 kcal/mol	PHE1808 (2.29 Å) PHE1801 (2.69 Å) ILE1809 (2.88 Å) GLU1810 (3.38 Å)	C-terminal / linker region ✗ PHE1760 pocket inaccessible in static conformation Off-target binding
holoMutant neutral MD rep (ligand removed)	−6.2 kcal/mol	ASN932 (1.76 Å) SER246 (2.00 Å) ALA242 (2.08 Å) TYR416 (2.69 Å)	Domain I/II surface ✗ Ligand-removed conformation displaces drug to non-functional region
WT static PDB (8VYK)	−5.3 kcal/mol	GLN90 (2.11 Å) THR89 (2.58 Å) LYS91 (2.76 Å) PHE93 (3.85 Å)	N-terminal loop ✗ PHE1760 pocket inaccessible in 8VYK conformation Off-target binding

Table 4. Blind docking Mode 1 closest residues across four receptor structures.

Only the apoMutant MD representative frame correctly directs protonated mexiletine to Domain IV S6 with PHE1760. This convergence with fpocket analysis confirms that the MD-derived apoMutant conformation is the appropriate receptor for mexiletine docking in the N347K system. All static or ligand-removed conformations produce off-target poses in non-functional surface regions despite comparable or better Vina scores.

> Higher Vina scores in static structures (N347K static: −6.5 vs apoMutant MD: −6.3) do not indicate better binding the poses are at biologically irrelevant sites. This illustrates the importance of residue-level pose analysis in addition to docking scores.

5. Targeted Docking: Pocket 2 vs Pocket 13

Using the apoMutant MD representative frame as receptor, targeted docking was performed at the two pharmacologically relevant sites identified by fpocket. Box centers were computed from fpocket vertex files (_vert.pqr) for each pocket. Box size: 25×25×25 Å, AutoDock Vina exhaustiveness = 128, 20 modes.

Site	Best Score	Pose Stability	fpocket Drugg.	Key Residues
Pocket 2 Domain IV S6 (classical LA site)	**−6.3 kcal/mol ★**	**Modes 1–11 all within 2–3 Å Very stable**	0.970	PHE1760, TYR1409, THR1461, LEU1462, SER1458, TRP1345, ILE1756, ILE1757, PHE1465
Pocket 13 Domain III/IV junction (protonated mex MD site)	−5.4 kcal/mol	Modes 1–4 within 2 Å Modes 5+ scattered	0.873	ILE1630, LEU1627, LEU1634, CYS260, LEU256, ILE1633, ILE1637, LEU1647

Table 5. Targeted docking results at Pocket 2 and Pocket 13 using apoMutant MD representative frame.

Pocket 2 (Domain IV S6) is the preferred site in targeted docking stronger score and much higher pose stability. The presence of PHE1760 and TYR1409 confirms the classical local anaesthetic mechanism (π–π interaction with the aromatic ring of mexiletine).

Pocket 13 shows lower docking affinity despite being the site visited by protonated mexiletine during 500 ns MD simulation (MM-GBSA $\Delta G = -15.01$ kcal/mol). This discrepancy reflects a fundamental limitation of docking: the rigid receptor cannot capture the cationic interactions with ASP1714 and ASP1423 that drive mexiletine to this site under physiological conditions. Docking systematically underestimates electrostatically-driven binding, especially for charged ligands in flexible environments.

6. Comprehensive Binding Affinity Summary

Table 6 consolidates all computational binding affinity data for protonated mexiletine in the Nav1.5 N347K system, integrating docking scores with MD-derived MM-GBSA values.

Method	Receptor	Score (kcal/mol)	Site	Notes
Blind docking Vina exh=128	N347K static PDB	−6.5	Off-target (C-terminal)	PHE1808 region. PHE1760 pocket inaccessible in static conformation. Score not biologically meaningful.
Blind docking Vina exh=128	apoMutant MD rep (index 3225)	**−6.3**	**Domain IV S6 PHE1760 ✔**	PHE1760 at 2.97 Å. Modes 1–5 converge at same site. Only MD-derived structure gives correct binding site in blind docking.
Blind docking Vina exh=128	holoMutant neutral MD rep (lig. removed)	−6.2	Off-target (Domain I/II)	ASN932, SER246. Ligand-removed conformation displaced — off-target.
Blind docking Vina exh=128	WT static PDB (8VYK)	−5.3	Off-target (N-terminal)	GLN90, THR89. PHE1760 pocket inaccessible in 8VYK conformation.
Targeted docking Pocket 2 Vina exh=128	apoMutant MD rep (index 3225)	**−6.3 ★**	**Domain IV S6 PHE1760 ✔**	Best targeted score. Modes 1–11 within 3 Å. PHE1760, TYR1409. Consistent with classical LA mechanism.
Targeted docking Pocket 13 Vina exh=128	apoMutant MD rep (index 3225)	−5.4	Domain III/IV junction	Lower docking score but confirmed by 500 ns MD. Docking underestimates cationic ASP1714/ASP1423 interactions.

Method	Receptor	Score (kcal/mol)	Site	Notes
MM-GBSA traj. average	holoMutant neutral mex 500 ns	**−18.47 ±3.81**	P-loop (neutral form)	Neutral mexiletine in P-loop. 6.5× selectivity vs WT (−2.84 kcal/mol). Note: neutral form is not physiological at pH 7.4.
MM-GBSA traj. average	holoMutant protonated mex 500 ns	**−15.01 ±5.59**	**Domain III/IV junction ✔**	Protonated mexiletine at Domain III/IV junction. ASP1423 Cationic 65.4%, ASP1714 Cationic 21.9%. Physiologically relevant binding mode.
MM-GBSA traj. average	holoWT neutral mex 500 ns	−2.84 ±5.10	P-loop (WT, neutral)	WT reference. Much weaker than N347K — confirms 6.5× mutation-selective effect in neutral form.
MM-GBSA traj. average	holoWT protonated mex (ongoing)	⏳ pending	TBD	holoWT protonated MD running (~1 week). Critical for ΔΔG = ΔG(N347K) − ΔG(WT) comparison at Domain III/IV junction.

Table 6. All binding affinity data. Blue = docking (apoMutant MD rep), Green = MM-GBSA, Yellow = pending.

7. Conclusion

7.1. Best binding site for drug design

Convergent evidence from fpocket, blind docking, targeted docking, and MD simulation identifies two pharmacologically relevant binding sites:

- Site 1 — Domain IV S6 (PHE1760): primary binding site confirmed by all methods. fpocket druggability 0.970–1.000 (MD structures), PHE1760 present, blind docking Mode 1 at PHE1760 (2.97 Å), targeted docking −6.3 kcal/mol with highest pose stability, consistent with classical local anaesthetic mechanism (Nakajima 2019).
- Site 2 — Domain III/IV junction (ASP1714, ASP1423): physiologically relevant site for protonated mexiletine. Confirmed by 500 ns MD simulation (ΔG = −15.01 kcal/mol, 95.2% occupancy), fpocket druggability 0.873. Docking underestimates this site due to rigid receptor limitations — MD is the appropriate method here.

7.2. Key methodological finding

Static crystallographic structures (both WT and N347K) are insufficient for binding site analysis in this system. In silico mutagenesis changes only one residue and does not capture the conformational consequences of $K347^{+}$. MD simulation is required to open the PHE1760-containing pocket to full druggability. This is demonstrated independently by fpocket (PHE1760 absent in static top pocket) and blind docking (off-target poses in static structures). The MD-derived apoMutant representative frame is validated as the correct receptor by both approaches.

7.3. Pending

holoWT protonated MD simulation is currently running (≈1 week remaining). Once complete:

- MM-GBSA ΔG for WT at Domain III/IV junction will be calculated
- ΔΔG = ΔG(N347K) − ΔG(WT) will quantify mutation-selective binding
- If WT shows weaker binding at Domain III/IV junction, this confirms N347K-specific preference for this site
- General meeting (Peri + Alessio + Silvia) to be arranged once results are complete

References

Frosio, A. et al. (2023). Unravelling novel SCN5A mutations linked to Brugada syndrome. Int. J. Mol. Sci. 24, 15089.

Nakajima, T. et al. (2019). Mexiletine block of voltage-gated sodium channels. Mol. Pharmacol. 95, 236–246.

Tikhonov, D.B. & Zhorov, B.S. (2022). Molecular modeling of cardiac sodium channel with mexiletine. Front. Pharmacol. PMID: 36557159.

Selig, E.E. et al. (2025). Proc. Natl. Acad. Sci. U.S.A. 122, e2416181122.

Le Guilloux, V. et al. (2009). fpocket: An open source platform for ligand pocket detection. BMC Bioinformatics 10, 168.

Trott, O. & Olson, A.J. (2010). AutoDock Vina: improving the speed and accuracy of docking. J. Comput. Chem. 31, 455–461.